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DS4420: Data Mining & Machine Learning 2

Building a Risk-Return Calculator with Long Short-Term Model (LSTM) and Bayesian Modelling

Purpose

- To help investors make an informed decisions based on the predicted return and the volatility of a company.
- To study if advanced machine learning models can capture the changes in the sectors.

Background & Data

- Financial markets are notoriously difficult to predict because of their non-linear and unstable nature, influenced by economic, political, and other factors.
- Set of 404 companies from the S&P500 from all major sectors with rich historical data dating back to 2005.

FEATURE	EXPLANATION
Risk-Free Rate	Return on a zero-risk investment (e.g., Treasury bonds); baseline
Inflation Rate	Rise in prices; reduces real returns
30-Day Rolling Average	Smoothed average price over 30 days; shows trends
S&P 500 Return	Market performance benchmark.
Company Return	Individual stock's performance
S&P 500 Std Dev	Market volatility
Company Std Dev	Stock-specific risk/volatility.

- Sector Distribution
    - Financials, Industrials, Technology, Healthcare ~14% each
    - All other sectors less than 10%

Methods

Long Short-Term Model (LSTM)

Learns the behaviour of historical sequential data such as stock returns to capture long-term dependencies.

Model = Sequential([
LSTM(100, return_sequences = TRUE),
LSTM(100),
Dense(50, activation = 'tanh'),
Dense(1)])

Tanh Activation (outputs between -1 & 1) because returns are often normally distributed

- No dropout because of the smaller monthly dataset size

Bayesian Modelling

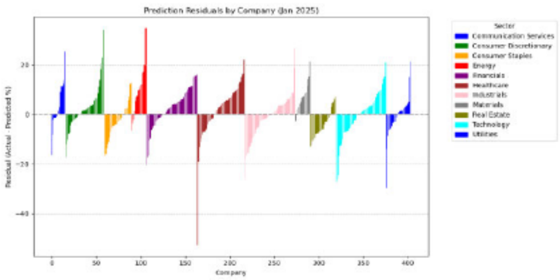
Modeled an Inverse Gamma-Gamma Distribution to estimate the variance of a stock.

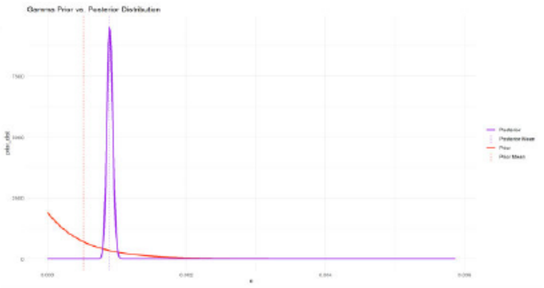
Likelihood $p(x_i \mid \theta)$	Model parameters	Conjugate Prior/Posterior $p(\theta \mid \Theta)$/ $p(\theta \mid x, \Theta) = p(\theta \mid \Theta')$	Prior Hyperparameters (Θ)	Posterior Hyperparameters $(\Theta)')$
Inverse Gamma with known α	β(inverse scale)	Gamma	α_0, β_0	$\alpha_0 + n\alpha$, $\beta_0 + \sum_{i=1}^{n} \frac{1}{x_i}$

Table 1. Conjugate Prior Table According to Wikipedia

Prior Mean = mean(Stocks in same sector)
Alpha Prior = 1 (uncertainty in initial guess)
Alpha Likelihood = 2 (modest confidence in data)

Results





- The mean squared error of the LSTM is 0.91% for January 2025 Predictions. Correlation and MSE is relatively high indicating poor predictions of returns and trends of whether stock will go up or down.
- There is a strong influence of the observed variances which causes a significant difference between the prior and posterior estimates/means.

Conclusions

- The LSTM perform poorly with macroeconomic indicators such as risk-free rate, inflation rate, and derived attribute such as 30-day rolling average, and standard deviation for stock return prediction.
- The Bayesian Model performed well under the assumptions that the stock's variance depends on
    - the variance of all stocks within the same sector
    - the variance of a stock is within a similar price range of the same sector

Future Work

- More qualitative and other quantitative sector-specific information is needed for increased accuracy.
- Focus on daily returns as opposed to monthly data to improve dataset size and increase the training dataset.
- Use more advance techniques such as a supervised Bayesian machine learning to evaluate how other factors affect variance.

References

Qiao, R., Chen, W., & Qiao, Y. (2022). Prediction of stock return by LSTM Neural Network. *Applied Artificial Intelligence*, *36*(1). https://doi.org/10.1080/08839514.2022.2151159